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SUB-SOIL INVESTIGATION REPORT

FOR PROPOSED
CONSTRUCTION OF WATER TANK
AT ARSENIC RISK REDUCTION PROJECT, KANAIGHAT, SYLHET.

CLIENT

EXECUTIVE ENGINEER, PWD, SYLHET.

Verification Code: 2yk6CeC4
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CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-100 ft	20
02	Disturbed sample	02	0-100 ft	20
03	Disturbed sample	03	0-100 ft	20
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

<u>Bearing capacity for deep foundation</u>

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a = Allowable bearing capacity, q_u = Unconfined Compressive Strength, γ = Unit Weight of soil, D = Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult} = Ultimate bearing capacity in kPa, N = SPT Number, FS = Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation = 9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Tables 7-11**.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5) **

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square (Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	3	0.302	0.317	0.346	0.356	0.385	0.419	0.325
	5.0-10	2	0.268	0.282	0.307	0.317	0.342	0.372	0.289
BH02	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH03	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH04	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH05	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

**** Design engineer should consider the settlement issue along with the tabulated bearing capacity for designing the shallow foundation.**

Table 8: Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.074	0.074	0.074	-	-	-	-	-	-	-
10	0.074	0.049	0.049	-	-	-	-	-	-	-
15	0.049	0.049	0.049	-	-	-	-	-	-	-
20	0.049	0.074	0.074	-	-	-	-	-	-	-
25	0.074	0.049	0.049	-	-	-	-	-	-	-
30	0.096	0.096	0.049	-	-	-	-	-	-	-
35	0.118	0.074	0.074	-	-	-	-	-	-	-
40	0.118	0.096	0.049	-	-	-	-	-	-	-
45	0.156	0.074	0.049	-	-	-	-	-	-	-
50	0.118	0.096	0.118	-	-	-	-	-	-	-
55	0.156	0.049	0.049	-	-	-	-	-	-	-
60	0.196	0.096	0.074	-	-	-	-	-	-	-
65	0.239	0.118	0.049	-	-	-	-	-	-	-
70	0.246	0.074	0.096	-	-	-	-	-	-	-
75	0.294	0.226	0.218	-	-	-	-	-	-	-
80	0.482	0.321	0.240	-	-	-	-	-	-	-
85	0.226	0.218	0.196	-	-	-	-	-	-	-
90	0.281	0.361	0.294	-	-	-	-	-	-	-
95	0.401	0.482	0.428	-	-	-	-	-	-	-
100	0.669	0.669	0.669	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	0.675	0.675	-	-	-	-	-	-	-
10	0.675	0.450	0.450	-	-	-	-	-	-	-
15	0.450	0.450	0.450	-	-	-	-	-	-	-
20	0.450	0.675	0.675	-	-	-	-	-	-	-
25	0.675	0.450	0.450	-	-	-	-	-	-	-
30	0.900	0.900	0.450	-	-	-	-	-	-	-
35	1.125	0.675	0.675	-	-	-	-	-	-	-
40	1.125	0.900	0.450	-	-	-	-	-	-	-
45	1.575	0.675	0.450	-	-	-	-	-	-	-
50	1.125	0.900	1.125	-	-	-	-	-	-	-
55	1.575	0.450	0.450	-	-	-	-	-	-	-
60	2.025	0.900	0.675	-	-	-	-	-	-	-
65	2.475	1.125	0.450	-	-	-	-	-	-	-
70	3.375	0.675	0.900	-	-	-	-	-	-	-
75	4.950	3.600	2.250	-	-	-	-	-	-	-
80	8.100	5.400	3.825	-	-	-	-	-	-	-
85	3.600	2.250	2.025	-	-	-	-	-	-	-
90	4.725	6.075	4.950	-	-	-	-	-	-	-
95	6.750	8.100	7.200	-	-	-	-	-	-	-
100	11.250	11.250	11.250	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.048	0.048	-	-	-	-	-	-	-
10	0.048	0.032	0.032	-	-	-	-	-	-	-
15	0.032	0.032	0.032	-	-	-	-	-	-	-
20	0.032	0.048	0.048	-	-	-	-	-	-	-
25	0.048	0.032	0.032	-	-	-	-	-	-	-
30	0.062	0.062	0.032	-	-	-	-	-	-	-
35	0.077	0.048	0.048	-	-	-	-	-	-	-
40	0.077	0.062	0.032	-	-	-	-	-	-	-
45	0.101	0.048	0.032	-	-	-	-	-	-	-
50	0.077	0.062	0.077	-	-	-	-	-	-	-
55	0.101	0.032	0.032	-	-	-	-	-	-	-
60	0.127	0.062	0.048	-	-	-	-	-	-	-
65	0.156	0.077	0.032	-	-	-	-	-	-	-
70	0.160	0.048	0.062	-	-	-	-	-	-	-
75	0.191	0.147	0.141	-	-	-	-	-	-	-
80	0.313	0.209	0.156	-	-	-	-	-	-	-
85	0.147	0.141	0.127	-	-	-	-	-	-	-
90	0.183	0.235	0.191	-	-	-	-	-	-	-
95	0.261	0.313	0.278	-	-	-	-	-	-	-
100	0.435	0.435	0.435	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	0.304	0.304	-	-	-	-	-	-	-
10	0.304	0.203	0.203	-	-	-	-	-	-	-
15	0.203	0.203	0.203	-	-	-	-	-	-	-
20	0.203	0.304	0.304	-	-	-	-	-	-	-
25	0.304	0.203	0.203	-	-	-	-	-	-	-
30	0.405	0.405	0.203	-	-	-	-	-	-	-
35	0.506	0.304	0.304	-	-	-	-	-	-	-
40	0.506	0.405	0.203	-	-	-	-	-	-	-
45	0.709	0.304	0.203	-	-	-	-	-	-	-
50	0.506	0.405	0.506	-	-	-	-	-	-	-
55	0.709	0.203	0.203	-	-	-	-	-	-	-
60	0.911	0.405	0.304	-	-	-	-	-	-	-
65	1.114	0.506	0.203	-	-	-	-	-	-	-
70	1.519	0.304	0.405	-	-	-	-	-	-	-
75	2.228	1.620	1.013	-	-	-	-	-	-	-
80	3.645	2.430	1.721	-	-	-	-	-	-	-
85	1.620	1.013	0.911	-	-	-	-	-	-	-
90	2.126	2.734	2.228	-	-	-	-	-	-	-
95	3.038	3.645	3.240	-	-	-	-	-	-	-
100	5.063	5.063	5.063	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed water tank construction.

Soil profile scenario

Overall soil information of the investigated site is little irregular in between the boreholes, and vertical inconsistency exists in the soil formation. Grey soft clay soil and grey stiff clay with silt with poor bearing capacity exist up to the top 60 ft in the boreholes 01, 02, and 03. Then, grey fine sand with silt and clay with moderate to good bearing capacity has been observed in the boreholes 01, 02, and 03.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Deep foundation (Bored/precast pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found from the attached table (8 to 11) in the report which is based on existing soil conditions.

Safety provisions

From the practical viewpoint, it has been found in many buildings that the structure failed due to soil settlement or judging the soil parameters incorrectly, providing no signs of structural damage. The earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of the foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix.

Special notes

These suggestions are absolutely tentative and the selection of type, depth, and dimensions of foundation would be decided by the Geotechnical Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of 100 ft as the client provided.

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